Knowledge Cutoff Effects on LLM-Generated Literature Reviews in Rapidly Evolving Fields

A Review of Temporal Limitations, Failure Modes, and Mitigation Strategies

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Large language models (LLMs) are increasingly used to accelerate literature review, a task made ever more burdensome by the exponential growth of scientific publication [11, 15]. Yet every LLM carries a fixed knowledge cutoff, and the gap between that cutoff and the true state of a fast-moving field introduces a distinctive class of error that is not well captured by general treatments of hallucination. This paper synthesizes recent empirical and methodological literature on knowledge cutoffs, temporal misalignment, and reference fabrication to characterize how these limitations manifest specifically in LLM-generated literature reviews. We show that reported cutoff dates are frequently inconsistent with a model's "effective" cutoff [4, 13], that models systematically blend pre- and post-cutoff knowledge in ways that are difficult for users to detect [21], and that citation fabrication rates in LLM-authored reviews range from roughly 30% to over 90% depending on model, domain, and topic maturity [2, 17]. We review current mitigation approaches—retrieval-augmented generation, knowledge editing, continual pretraining, and prompt-based temporal grounding—and assess their differential effectiveness for rapidly evolving domains such as artificial intelligence, biomedicine, and law, where the interval between discovery and consolidation into training corpora is shortest. We conclude by identifying open research gaps, including the absence of standardized cutoff-awareness benchmarks specific to review generation, the need for domain-adaptive retrieval that privileges recency without sacrificing precision, and the lack of longitudinal evaluation frameworks that track review quality as fields evolve after a model's training snapshot.

Additional Key Words and Phrases: large language models, knowledge cutoff, temporal misalignment, literature review automation, hallucination, retrieval-augmented generation, knowledge editing, citation fabrication, rapidly evolving fields, systematic review

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1 Introduction

The literature review is the connective tissue of scientific work: it situates a contribution relative to what is already known, and its quality depends directly on how completely and accurately it represents the current state of a field. Large language models have been proposed, and increasingly adopted, as tools to accelerate this labor-intensive process, automating search-string construction, title-and-abstract screening, data extraction, and synthesis [16, 19]. Systematic reviews of the review-automation literature report that hundreds of projects have applied LLMs to one or more stages of the review pipeline, with GPT-based architectures dominant and data extraction the most mature use case [16]. Every such application, however, inherits a structural property of the underlying model: its knowledge is frozen at a

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training-data cutoff, after which the model has no direct exposure to new publications, retracted findings, superseded methods, or terminology shifts [4]. This limitation is not merely cosmetic. Unlike a human reviewer, who can consciously flag "I have not read anything published after date X," an LLM will typically generate fluent, confident prose about a field's state of the art regardless of whether that state of the art falls before or after its cutoff [10]. The result is a review that reads as comprehensive and current but may in fact be silently anchored to a snapshot of the literature that is months or years stale — a failure mode that is especially consequential in domains where the literature itself evolves on a timescale comparable to, or faster than, the interval between model releases. This problem is compounded by a second, related phenomenon: LLMs asked to generate citations for a literature review frequently fabricate references outright, inventing plausible-sounding titles, author lists, and venues for papers that do not exist [2, 17]. Cutoff effects and citation hallucination interact in a specific and underexamined way: a model may correctly refuse or hedge when asked directly about an event after its cutoff, yet still confidently cite a fabricated "recent" paper in a synthesized review, because narrative synthesis engages different generation dynamics than direct factual recall [10, 21]. This paper reviews the state of research on knowledge cutoffs as they specifically bear on LLM-generated literature reviews in rapidly evolving fields — domains such as artificial intelligence, genomic medicine, and public-health emergency response, where the doubling time of the literature can be measured in single-digit years rather than decades [11, 15]. We organize the discussion around four questions: (1) What do we know empirically about how LLM knowledge cutoffs actually behave, as distinct from what providers report? (2) How do these cutoff effects manifest as concrete failure modes in review generation? (3) What mitigation strategies exist, and how well do they perform specifically for fast-moving domains? (4) What gaps remain in evaluating and addressing this problem? *Document structure.* The discussion first establishes the scale and temporal boundary problem (Sections 2.1 and 2.2), then examines the resulting failure modes (Section 3) and mitigation approaches (Section 4).

2 Background and Related Work

2.1 The Growth of the Literature LLMs Are Asked to Review

The motivation for LLM-assisted review is inseparable from the scale of contemporary scientific output. Global science and engineering article output reached approximately 3.3 million in 2022, up from 2.0 million in 2010, an overall growth rate on the order of 4–5% annually [15]. In computation- and AI-adjacent fields the growth is markedly steeper: arXiv itself passed three million cumulative submissions within four years of reaching two million, compared with the eight years the platform took to grow from one million to two million (cited in recent literature-synthesis tooling work) [11, 19]. This acceleration is precisely what makes automated review attractive and, simultaneously, what makes any fixed-cutoff model's coverage of a field decay fastest in the domains where reviews are most needed. The literature map used for the review is shown in Figure 1.

2.2 Knowledge Cutoffs: Reported versus Effective

A model's publicly reported cutoff date is not a reliable proxy for what the model actually knows. [4] introduce the notion of an "effective cutoff," distinct from the reported cutoff and variable across sub-resources and topics, and show empirically that effective cutoffs diverge from reported ones because of temporal biases in web-crawl data (older content persists disproportionately in newer crawls) and imperfect deduplication across near-duplicate documents. Building on this, [13] introduce LLMLagBench, a freshness benchmark that estimates the earliest probable temporal boundary of a model's training data from its knowledge of recent events, and find that several evaluated models exhibit multiple

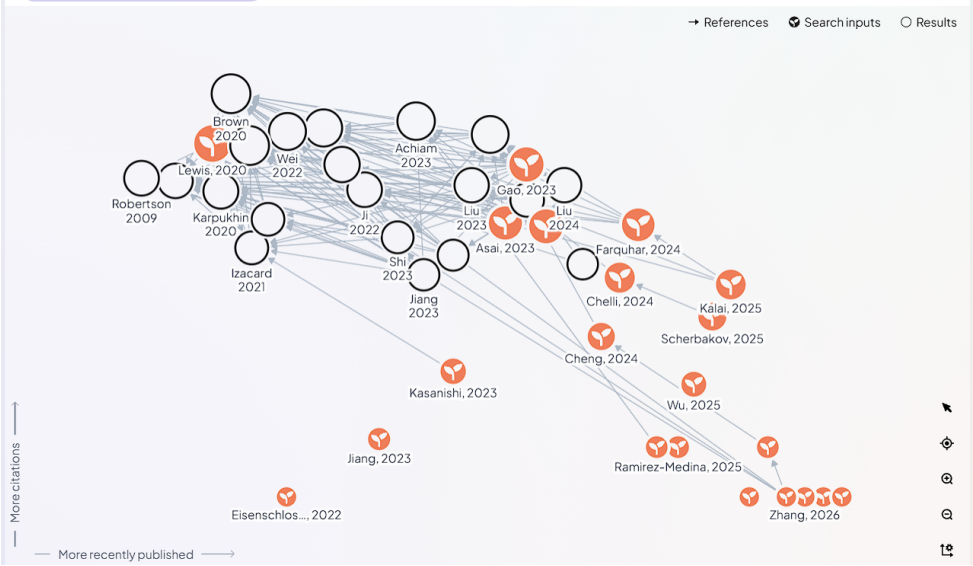


Fig. 1. ResearchRabbit literature map used in the review.

partial cutoff points, plausibly corresponding to distinct pretraining and post-training stages; strikingly, at least one open model's self-reported cutoff was nearly two years earlier than its benchmark-estimated boundary, indicating that models can be simultaneously overconservative in self-report and prone to hallucinating about genuinely post-cutoff events.

2.3 Temporal Misalignment and Knowledge Conflict

A parallel line of work examines how models handle facts that change over time. [5] formalize the notion of time-aware language modeling, showing that models pretrained on temporally heterogeneous corpora can be probed for what was true at a given point versus what is true "now," and that models often default to earlier-learned associations even when later information is available. More recent work using the WIKIRECENTCHANGES benchmark shows that models spontaneously produce explicit temporal reasoning when a queried fact has actually changed since their cutoff, but that this reasoning rarely propagates into the model's final answer — models "know" a fact may be stale in an intermediate reasoning trace yet still assert the stale version in the response users see (cited work on temporal knowledge conflict resolution, 2026) [5]. [21] further show that apparent post-cutoff performance decay, often used as a signal of benchmark contamination, is itself highly sensitive to how questions are phrased, complicating efforts to cleanly separate "the model doesn't know this" from "the model was tested in a way that leaked prior knowledge."

2.4 Hallucination and Citation Fabrication in Review Generation Specifically

Independent of temporal effects, LLMs asked to produce scholarly references are documented to fabricate them at high rates [8]. [2] compared ChatGPT-3.5, GPT-4, and Bard on reproducing systematic-review reference lists in orthopedics and found hallucination rates of 39.6%, 28.6%, and 91.4% respectively, with precision (the fraction of generated references that were genuine and relevant) below 15% for all three models. [3, 17] introduce a multidimensional framework for

evaluating LLM-generated literature reviews across reference generation, abstract writing, and full review composition, and report that even the most advanced contemporary models continue to hallucinate references, with performance varying substantially by academic discipline. Hallucination-detection work also shows that uncertainty can be assessed from semantic variation in model outputs rather than only by checking individual references after generation [6]. A within-domain study of mental-health topics found that citation fabrication rates depend on topic familiarity and prompt specificity, with less publicly visible or less scientifically mature subtopics eliciting substantially higher fabrication rates than well-covered ones (cited PMC study on citation fabrication, 2025), a finding directly relevant to rapidly evolving subfields where public visibility lags genuine research activity.

2.5 Automated Review-Generation Systems

A growing body of systems work builds pipelines intended to mitigate these problems by construction. [16] systematically reviewed 172 studies on LLM-assisted review automation and found that GPT-based models achieved a mean data-extraction precision of 83.0% and recall of 86.0%, but that publication search remained the weakest stage, attributable specifically to citation hallucination when models were prompted to generate references directly rather than retrieve them. [19] describe an end-to-end pipeline that grounds every generated statement in retrieved source text and reports hallucination risk below 0.5% at 95% confidence through multilayered quality control and full data-stream traceability. [11] released SciReviewGen, a large-scale dataset of over 10,000 human-written literature reviews and 690,000 cited papers, enabling supervised and retrieval-grounded approaches to review generation and providing an empirical basis for measuring hallucination and information completeness against a human-authored reference standard.

3 Main Thematic Discussion: How Cutoff Effects Manifest in Review Generation

3.1 Silent Staleness

The most consequential failure mode for rapidly evolving fields is silent staleness: a review that omits an entire wave of post-cutoff work without any signal to the reader that omission has occurred. Because LLMs generate review prose in the same confident, well-formed register regardless of the recency of their underlying knowledge, a reader cannot distinguish a comprehensive account of a field from an account that is comprehensive only up to a training cutoff that may be a year or more in the past [4, 13]. In domains with multi-year knowledge doubling times this is a manageable degradation; in AI, genomics, or outbreak epidemiology, where a training cutoff of even six months can mean the omission of a field-defining architecture, variant, or trial result, silent staleness can materially mislead downstream research prioritization.

3.2 Blended and Contradictory Temporal Knowledge

A second manifestation is the blending of pre- and post-cutoff information within a single generated passage. LLMLag-Bench's finding of multiple partial cutoff points within individual models [13] implies that different facts within the same review may be drawn from different effective temporal horizons, so that a synthesized narrative can, without any internal inconsistency signal, combine an accurate account of foundational work with an inaccurate or hallucinated account of "recent" developments. This is distinct from — and arguably harder to detect than — a uniformly stale review, because the presence of some genuinely current-sounding content lends false credibility to adjacent fabricated content.

3.3 Citation Fabrication as a Distinct but Interacting Failure

Citation fabrication is not caused by knowledge cutoffs per se — models hallucinate references even for well-established, pre-cutoff topics [2] — but cutoff pressure exacerbates it in a specific way. When a model is asked to review "recent" or "current" work in a fast-moving field, it faces a structural incentive to produce content that sounds temporally current even where its training data thins out near the cutoff boundary; the topic-familiarity effect documented for mental-health subtopics (citation fabrication study, 2025) suggests that the same dynamic will be sharper for genuinely emergent subfields, where the sparse, recent, or preprint-heavy literature that does exist in training data is exactly the material a model is least well calibrated to recall precisely.

3.4 Benchmark Contamination and the Limits of Temporal Self-Evaluation

A further complication is that measuring a model's temporal knowledge boundary is itself methodologically fraught. [21] show that whether a benchmark question exhibits the expected "post-cutoff performance decay" signal depends heavily on how the question is constructed from source material, even when the underlying facts are held constant; simple transformations of the same underlying questions can eliminate an otherwise clear contamination signal. This matters directly for literature-review evaluation because it undermines naive approaches to auditing review currency (e.g., simply asking a model what it knows about a topic as of a given date) as a reliable diagnostic.

4 Current Approaches and Developments

4.1 Retrieval-Augmented Generation

Retrieval-augmented generation (RAG) remains the most widely adopted mitigation for both staleness and citation fabrication [7, 12]. By grounding generation in retrieved passages from an external, updatable corpus rather than relying solely on parametric memory, RAG systems can in principle surface literature published after a model's training cutoff and provide a verifiable textual basis for each citation. Active and adaptive retrieval variants extend this further by allowing the model to decide when and what to retrieve mid-generation rather than retrieving once up front [9], and self-reflective approaches such as Self-RAG train models to critique their own retrieval decisions and generated claims [1]. For review generation specifically, systems such as those described by [19] report hallucination rates below 0.5% when every generated claim is text-restricted to retrieved source material with full traceability back to a DOI. However, RAG's effectiveness for rapidly evolving fields is bounded by the currency and completeness of the retrieval corpus itself: a RAG system connected to a database that lags the literature by weeks still inherits a cutoff, merely a more recent one, and retrieval quality studies show that noisy or poorly ranked retrieved passages can degrade rather than improve output reliability (surveyed in recent RAG-hallucination literature).

4.2 Knowledge Editing and Continual Pretraining

A second family of approaches modifies the model's parametric knowledge directly rather than supplementing it externally. Knowledge editing techniques such as ROME, MEMIT, and their lifelong-editing successors (e.g., GRACE and WISE) aim to insert or correct discrete facts without full retraining, using external memorization, global optimization, or local parameter modification strategies [18]. These methods are well suited to correcting discrete, well-defined factual errors (e.g., an organization's leadership, a superseded classification) but are less naturally suited to the kind of broad, unstructured currency required for literature review, where "new knowledge" typically arrives as entire papers or subfields rather than as clean (subject, relation, object) triples. Continual learning approaches that periodically

re-pretrain or fine-tune models on newly published text address this scale mismatch but reintroduce the cost, retraining cadence, and catastrophic-forgetting concerns that motivated RAG and editing approaches in the first place [20].

4.3 Prompt-Based and Inference-Time Temporal Grounding

A lighter-weight class of methods attempts to improve temporal behavior through prompting or inference-time constraints alone, without touching model parameters or requiring external retrieval infrastructure. Explicit-cutoff prompting, in which a model is told its assumed reference date and instructed to reason only from information available before it, has been shown to reduce — though not eliminate — leakage of later knowledge into ex-ante reasoning tasks, with prefix-positioned constraints outperforming suffix-positioned ones (temporal-critique literature). Reward-based training approaches such as TEMPO explicitly penalize post-cutoff claim leakage as a hard constraint before optimizing for task performance, reducing measured leakage from a 2–13% baseline range to 0.6–3.7% across evaluated conditions [22]. These approaches are attractive for review generation because they require no infrastructure changes, but current evidence suggests they are more effective at suppressing overt post-cutoff assertions than at inducing genuine, generalizable temporal discipline across an open-ended synthesis task.

4.4 Domain-Specific and Legal/Regulatory Applications

Domain-specific evaluations illustrate both the promise and limits of current mitigations under realistic conditions. In statutory question answering, where legal texts are amended on well-defined dates, RAG variants that enforce temporal validity through fact-date extraction and version filtering substantially improved performance over an unconstrained baseline across all tested question types, while naïve web-search augmentation produced unstable gains and exhibited a marked recency bias — the tendency to prefer newer provisions even when an older, historically appropriate version governs the query [14]. This finding is instructive for literature review: it suggests that the naïve fix of "just add web search" is not sufficient and can introduce a new bias (favoring whatever is most recent) that is nearly as distorting as staleness itself, particularly for reviews that must correctly represent the historical development of a field rather than only its current state.

Table 1. Comparison of the main mitigation approaches discussed in this review.

Approach	Main idea	Main benefit	Main limitation
RAG	Retrieve external, updateable sources during generation	Improves currency and traceability	Depends on corpus freshness and retrieval quality
Knowledge editing	Modify selected model knowledge directly	Can correct discrete factual errors	Poor fit for broad, continuously growing literature
Continual pretraining	Periodically update model parameters with new data	Broadens parametric knowledge	Cost and catastrophic-forgetting risk
Temporal grounding	Prompt or constrain the model around a reference date	Reduces overt temporal leakage	Limited validation for open-ended review synthesis
Domain-aware retrieval	Combine retrieval with temporal/version constraints	Balances recency with historical validity	Requires domain-specific temporal policies

5 Research Challenges and Limitations

Several challenges recur across the literature reviewed above. The approaches and trade-offs summarized in Table 1 motivate these challenges. First, reported cutoff dates cannot be taken at face value: effective cutoffs vary by sub-resource

and topic, multiple partial cutoffs can coexist within a single model, and models can be simultaneously overconservative in self-reported cutoff and prone to hallucinating about events beyond their true boundary [4, 13]. Any evaluation of an LLM-generated review that relies on the provider's stated cutoff as ground truth for currency is therefore built on an unreliable premise. Second, benchmark and evaluation methodology for temporal behavior is itself fragile. The demonstration that post-cutoff performance decay — the standard signal used to detect contamination or measure temporal boundaries — is highly sensitive to how test questions are constructed from source material [21] means that claims about a model's temporal knowledge boundary, including many of the claims surveyed in this paper, should be read as benchmark-dependent rather than as intrinsic model properties. Third, mitigation techniques trade off currency against other desiderata in ways that are not yet well characterized for the review-generation task specifically. RAG improves currency but inherits the completeness and quality of its retrieval corpus and index freshness, and can be degraded by noisy retrieval [7]. Knowledge editing operates at the granularity of discrete facts, which is a poor match for the unstructured, large-scale nature of "a paper was published" as a unit of new knowledge [18]. Continual pretraining reintroduces cost and catastrophic-forgetting risk [20]. Prompt- and reward-based temporal grounding reduce overt leakage but have chiefly been validated on narrow factual or forecasting tasks rather than on open-ended synthesis [22], and it remains unclear how well these gains transfer to the qualitatively different task of narrative literature review. Fourth, citation fabrication and temporal staleness interact but are measured largely in separate literatures. Studies of hallucination rates in review generation [2, 17] do not typically decompose fabricated citations by whether the ostensible source would have fallen before or after the model's cutoff, and studies of temporal knowledge boundaries [4, 13] do not typically evaluate citation behavior. As a result, there is no existing empirical estimate of how much of the citation fabrication observed in LLM-generated reviews of fast-moving fields is specifically attributable to cutoff pressure as opposed to the baseline hallucination rate observed even for well-established topics. Fifth, and more broadly, the growth rate of the literature itself is not fixed across fields, and mitigation strategies validated on slower-moving domains (general knowledge QA, encyclopedic fact recall) may not generalize to domains where the interval between a result's appearance (often first as a preprint) and its consolidation into curated, retrievable corpora is itself short and variable — precisely the domains, such as frontier AI research or outbreak response, where up-to-date literature review is most valuable and most difficult to guarantee.

6 Research Gaps and Future Directions

Building on the challenges above, we identify four priority directions for future work. First, standardized, review-specific cutoff-awareness benchmarks are needed. Existing temporal-boundary benchmarks such as LLMLagBench [13] and dated-data probing methods [4] evaluate factual recall or event knowledge, not the review-synthesis task itself. A benchmark that pairs a fixed corpus of ground-truth literature reviews across several rapidly evolving subfields with known publication-date distributions would allow direct measurement of silent staleness, temporal blending, and citation fabrication within the same evaluation, rather than inferring review quality from proxy tasks. Second, domain-adaptive retrieval calibrated to a field's evolution rate deserves more attention. The legal-domain finding that naïve recency weighting introduces its own bias [14] suggests that effective retrieval augmentation for literature review needs field-aware temporal weighting — treating recency as informative but not dispositive, and differentiating between fields where the newest source is usually the most authoritative and fields (such as much of applied science) where historical context remains essential to accurate synthesis. Third, joint modeling of citation fabrication and temporal cutoff effects is an open empirical gap. Future studies could deliberately vary the ostensible recency of requested citations (asking a model to cite work from progressively later windows relative to its estimated effective

cutoff) to characterize whether and how fabrication rate increases as a function of proximity to, and distance beyond, the cutoff boundary, building on the topic-familiarity effects already documented (citation fabrication study, 2025). Fourth, longitudinal evaluation frameworks are needed that track a fixed model's review quality on a fixed topic as that topic's true literature continues to grow after the model's training snapshot — effectively measuring the half-life of an LLM-generated review's accuracy. This would connect naturally to continual-learning and knowledge-editing research [18, 20] by providing an outcome metric (review currency decay rate) against which the practical value of periodic model updates, targeted edits, or retrieval-corpus refresh cadences could be compared directly, rather than evaluated only on synthetic editing benchmarks. Finally, given that even the most careful current pipelines report residual hallucination rates in the fraction-of-a-percent to several-percent range under controlled, retrieval-grounded conditions [19], and substantially higher rates under naïve direct generation [2], an important practical direction is the development of user-facing currency and confidence indicators — surfacing, alongside a generated review, an estimate of the model's effective topical cutoff and a calibrated warning where synthesis relies on sparse or near-boundary training coverage, rather than presenting all generated review content with uniform apparent confidence.

7 Conclusion

Large language models offer real efficiency gains for literature review, particularly in structured stages such as screening and data extraction, where reported precision and recall are already competitive with human reviewers [16]. But their fixed knowledge cutoffs, combined with a documented tendency toward confident citation fabrication, create a specific and underappreciated risk for reviews of rapidly evolving fields: outputs that are fluent, well-organized, and superficially comprehensive, yet silently anchored to a training snapshot that may already be substantially out of date at the moment of use. The empirical literature makes clear that this risk cannot be managed by trusting a provider's reported cutoff date, since effective cutoffs are inconsistent, topic-dependent, and sometimes actively misrepresented by the model's own self-report [4, 13]. Current mitigations — retrieval augmentation, knowledge editing, continual pretraining, and prompt- or reward-based temporal grounding — each address part of the problem but none has yet been comprehensively validated on the open-ended synthesis task of review generation itself, particularly for domains where the underlying literature evolves fastest. Closing this gap will require benchmarks that evaluate cutoff-awareness within the review-generation task directly, retrieval methods that weight recency without overcorrecting toward it, and evaluation frameworks that treat review accuracy as something that decays over time rather than as a fixed, one-time property of a model's output. Until such tools mature, human verification of currency and citation authenticity remains an indispensable, non-optional component of any LLM-assisted literature review in a fast-moving field.

References

[1] Akari Asai, Zeqiu Wu, Yizhong Wang, Avi Sil, and Hannaneh Hajishirzi. 2024. Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection. In *Proceedings of the Twelfth International Conference on Learning Representations*.

[2] M. Chelli, J. Descamps, V. Lavoué, C. Trojani, M. Azar, M. Deckert, J.-L. Raynier, G. Clowez, P. Boileau, and C. Ruetsch-Chelli. 2024. Hallucination rates and reference accuracy of ChatGPT and Bard for systematic reviews: Comparative analysis. *Journal of Medical Internet Research* 26 (2024), e53164. doi:10.2196/53164

[3] X. Chen and X. Zhang. 2025. Large language models streamline automated systematic review: A preliminary study. *arXiv* (2025). arXiv:2502.15702 doi:10.48550/arXiv.2502.15702

[4] J. Cheng, M. Marone, O. Weller, D. Lawrie, D. Khashabi, and B. Van Durme. 2024. Dated data: Tracing knowledge cutoffs in large language models. *arXiv* (2024). arXiv:2403.12958 doi:10.48550/arXiv.2403.12958

[5] B. Dhingra, J. R. Cole, J. M. Eisenschlos, D. Gillick, J. Eisenstein, and W. W. Cohen. 2022. Time-aware language models as temporal knowledge bases. *Transactions of the Association for Computational Linguistics* 10 (2022), 257–273. doi:10.1162/tacl_a_00459

[6] S. Farquhar, J. Kossen, L. Kuhn, and Y. Gal. 2024. Detecting hallucinations in large language models using semantic entropy. *Nature* 630 (2024), 625–630. doi:10.1038/s41586-024-07421-0

[7] Y. Gao, Y. Xiong, X. Gao, K. Jia, J. Pan, Y. Bi, Y. Dai, J. Sun, and H. Wang. 2023. Retrieval-augmented generation for large language models: A survey. *arXiv* (2023). arXiv:2312.10997 doi:10.48550/arXiv.2312.10997

[8] L. Huang, W. Yu, W. Ma, W. Zhong, Z. Feng, H. Wang, Q. Chen, W. Peng, X. Feng, B. Qin, and T. Liu. 2025. A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions. *ACM Transactions on Information Systems* 43, 2 (2025), Article 42. doi:10.48550/arXiv.2311.05232

[9] Z. Jiang, F. F. Xu, L. Gao, Z. Sun, Q. Liu, J. Dwivedi-Yu, Y. Yang, J. Callan, and G. Neubig. 2023. Active retrieval augmented generation. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, 7969–7992.

[10] A. T. Kalai, O. Nachum, S. S. Vempala, and E. Zhang. 2025. Why language models hallucinate. *arXiv* (2025). arXiv:2509.04664 doi:10.48550/arXiv.2509.04664

[11] T. Kasanishi, M. Isonuma, J. Mori, and I. Sakata. 2023. SciReviewGen: A large-scale dataset for automatic literature review generation. *arXiv* (2023). arXiv:2305.15186 doi:10.48550/arXiv.2305.15186

[12] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W. Yih, T. Rocktäschel, S. Riedel, and D. Kiela. 2020. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems 33*. 9459–9474.

[13] P. Pęzik, K. Kaczyński, M. Szymańska, F. Żarnecki, Z. Deckert, J. Kwiatkowski, and W. Janowski. 2025. LLMLagBench: Identifying temporal training boundaries in large language models. *arXiv* (2025). arXiv:2511.12116 doi:10.48550/arXiv.2511.12116

[14] M. Prior, A. Schultz, and M. Grabmair. 2026. Asking for an old friend: Diagnosing and mitigating temporal failure modes in LLM-based statutory question answering. *arXiv* (2026). arXiv:2605.23497 doi:10.48550/arXiv.2605.23497

[15] J. Ramirez-Medina, M. Ataei, and A. Amirfazli. 2025. Accelerating scientific research through a multi-LLM framework. *arXiv* (2025). arXiv:2502.07960 doi:10.48550/arXiv.2502.07960

[16] D. Scherbakov, N. Hubig, V. Jansari, A. Bakumenko, and L. A. Lenert. 2025. The emergence of large language models as tools in literature reviews: A large language model-assisted systematic review. *Journal of the American Medical Informatics Association* 32, 6 (2025), 1071–1086. doi:10.1093/jamia/ocaf063

[17] X. Tang, X. Duan, and Z. G. Cai. 2025. Large language models for automated literature review: An evaluation of reference generation, abstract writing, and review composition. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*. arXiv:2412.13612 doi:10.48550/arXiv.2412.13612

[18] S. Wang, Y. Zhu, H. Liu, Z. Zheng, C. Chen, and J. Li. 2024. Knowledge editing for large language models: A survey. *Comput. Surveys* (2024). doi:10.1145/3698590

[19] S. Wu, X. Ma, D. Luo, L. Li, X. Shi, X. Chang, X. Lin, R. Luo, C. Pei, C. Du, Z.-J. Zhao, and J. Gong. 2025. Automated literature research and review-generation method based on large language models. *National Science Review* (2025). arXiv:2407.20906 doi:10.1093/nsr/nwaf169

[20] T. Wu, L. Luo, Y.-F. Li, S. Pan, T.-T. Vu, and G. Haffari. 2024. Continual learning for large language models: A survey. *arXiv* (2024). arXiv:2402.01364 doi:10.48550/arXiv.2402.01364

[21] T. J. Zhang, G. Dev, N. Wang, M. Obreiter, P. S. Pandey, K. Samway, W. Jiang, Y. Huang, B. Schölkopf, M. Sachan, and Z. Jin. 2025. Test of time: Rethinking temporal signal of benchmark contamination. *arXiv* (2025). arXiv:2509.00072 doi:10.48550/arXiv.2509.00072

[22] Z. Zhang and B. C. Stadie. 2026. TEMPO: Temporal enforcement via mode-separated policy optimization for trustworthy LLM backtesting. *arXiv* (2026). arXiv:2605.18843 doi:10.48550/arXiv.2605.18843